



RTL2GDSII WORKSHOP

**DESIGN COMPILER
(Synthesis Tool)**

LOGIC SYNTHESIS

Only for Chip to startup (C2S) Training



RTL2GDSII WORKSHOP

- Previous session : HDL Prog. & Simulation of 4-bit full adder design

Logic Synthesis :

- Converts a design description written in a HDL, such as Verilog or VHDL, into an optimized gate-level netlist mapped to a specific logic library.

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Logic Synthesis :- Includes 3 steps

Synthesis = Translation + Logic Optimization + Gate Mapping

- 1. Translation** : Conversion of High Level Description of design (**HDL**) into **Combinational Boolean Equations** and **Sequential Memory Elements**.
(Tool generates the **Gate level representation – Tech. Independent - GTECH**)
- 2. Logic Optimization** : Tool Optimizes the Boolean exp./Seq. Elements and **selects** a combination of **library cells** that meets the functional, timing, area, and power requirements of the design. (Tech. **Independent**)
- 3. Gate Mapping** : Tool maps the optimized equations and memory elements with **Technology dependent cell library** for design implementation.



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Steps : Synthesis Flow of Design Compiler :-

1. **Develop HDL files** – Creation of HDL files to describe design.
2. **Specify the libraries** – Link Lib, Target Lib, Symbol Lib, Synthetic Lib
3. **Read Design** – analyze and elaborate design
4. **Set the design constraints** – Design Rule and Optimization Constraints
5. **Synthesize and Optimize the design** – Compile (Mapping to Cell Library)
6. **Save the design database** – Write the mapped netlist

LINK LIB

- For a design to be complete , all **cell instances** in the design must be **linked to the library components**.
- This process is called **linking the design** or **resolving references**.
- The logic libraries that Design Compiler uses to **resolve cell references** are called **link libraries**.
- **Link libraries** define the **delay models** that are used to calculate timing values and path delays.

TARGET LIB

Target Lib :

- Contains the cells used **to generate the netlist** and definitions for designs operating conditions.
- Design Compiler selects **functionally correct gates** from the **target libraries** to build a circuit during mapping. It also calculates the **timing of the circuit** by using the vendor-supplied timing data for these gates.
- It is the **Logical library used for mapping** during optimization.
- The **target libraries are the subset of the link libraries** that are used to compile or translate a design.



Symbol Libraries (.sdb)

- **Symbol libraries** contains **graphical symbols** that represent library cells in design schematics.
- Design Compiler **uses symbol libraries to generate schematic views.**
- To generate a schematic, **Design Compiler performs a one-to-one mapping of cells in the netlist to cells in the symbol library.**
- Each Design Compiler installation includes a default symbol library file, **generic.sdb**, and this file contains **generic gate symbols.**
- If the specified libraries do not contain the symbols , the tool displays the **cell instance as a rectangle**, which is the **default representation.**



DesignWare/Synthetic libraries (.sldb)

- A DesignWare library is a **collection of reusable circuit-design** building blocks (components).
- DesignWare **components** implements many of the **built-in HDL operators** are provided by Synopsys.
- These operators include **+, -, *, <, >, <=, >=**, and the operations defined by if and case statements.
- During synthesis, Design Compiler **selects the component with the best speed and area optimization** from the DesignWare library.



DC command : analyze

The analyze command performs the following **tasks**:-

- Reads an HDL source file
- Checks for errors before building generic logic for the design
- Creates HDL library objects in an HDL-independent intermediate format
- Stores the intermediate files in a location you define

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DC command: elaborate

The elaborate command performs the following tasks :-

- Translates the design into a technology-independent design (**GTECH**) netlist from the intermediate files produced during analyze command execution.
- Replaces the HDL arithmetic operators in the code with DesignWare components.
- Automatically executes the link command, which resolves design references.

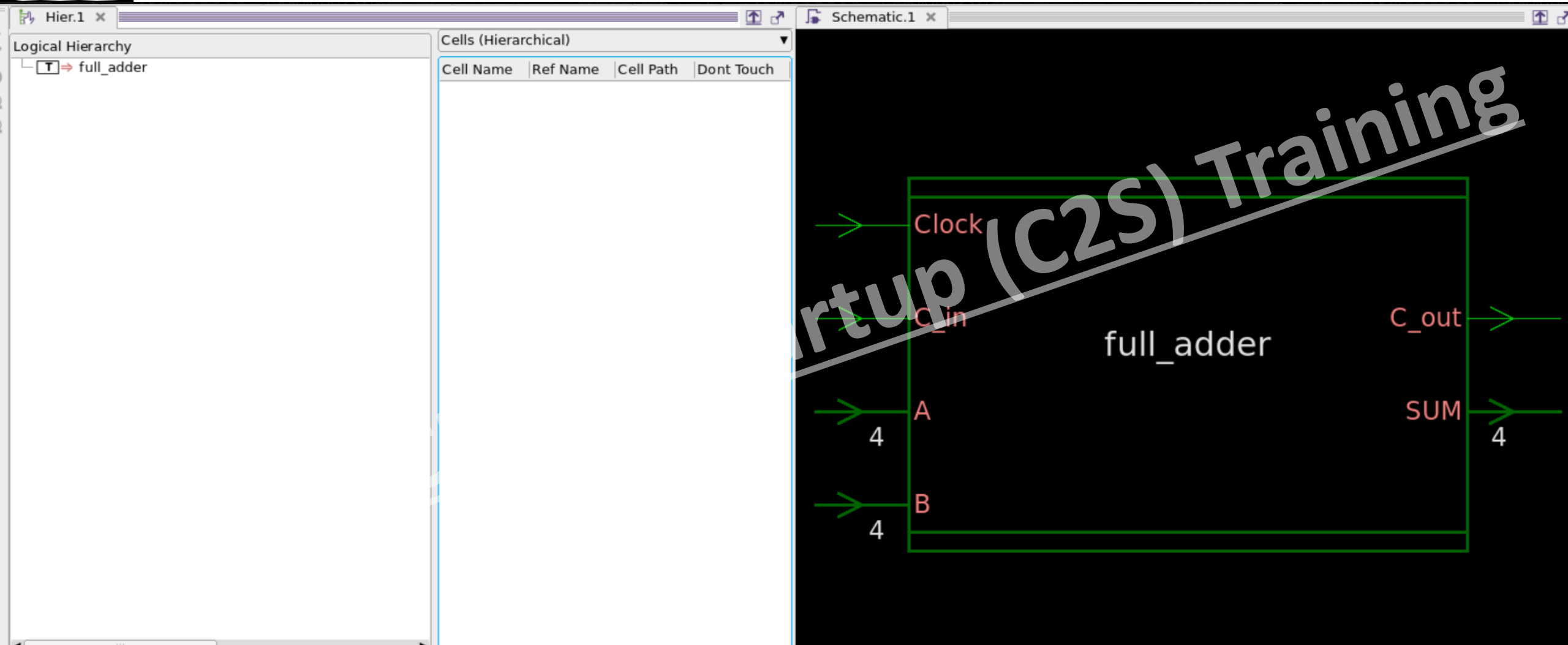


COMMANDS EXECUTION FOR LOGIC SYNTHESIS

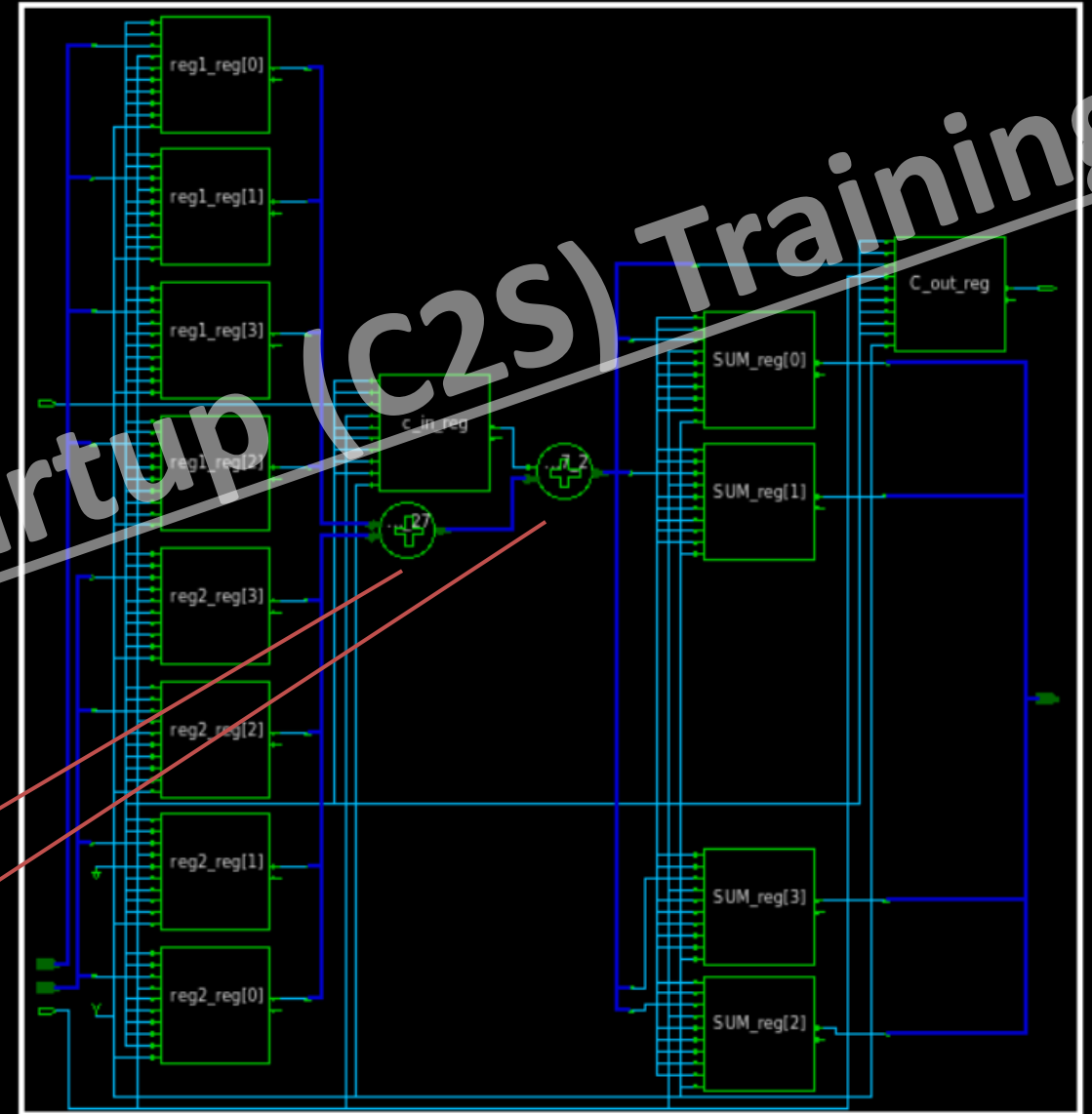
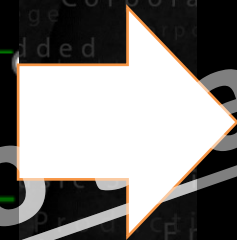
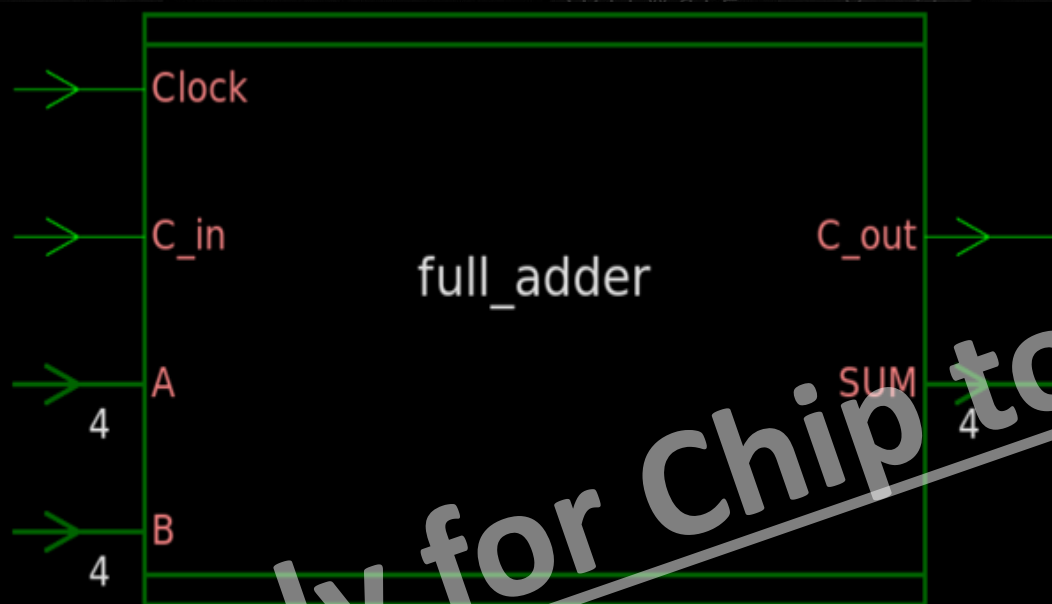
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ADD Operator Symbol

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